

A Clock That Cannot Be Dragged: Why the Preferred-Frame Test Does Not Bite a Cuscuton-Clock MOND Sector

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Abstract

A clock field that defines a preferred foliation is normally in danger from the post-Newtonian preferred-frame parameters: a body moving at velocity w through the clock's rest frame sees, in its own frame, a clock with a spatial gradient $\nabla'\tau = s_0 w$, and an order-unity coefficient in front of that tilt is excluded by roughly a factor of twelve for the solar system. An earlier action in this programme was killed on exactly this test. We show that a cuscuton-clock MOND sector of the form $P(X, \tau) - V(\tau) + sW(Y, \tau)$, with matter minimally coupled, is blind to the boost at background level, for a reason that is structural rather than numerical. Because the background scalar depends only on the clock, $\chi = \bar{\chi}(\tau)$, its gradient $\partial_\mu \bar{\chi} = q \partial_\mu \tau$ is parallel to the clock's unit normal; the MOND invariant Y is by construction the part of $\partial\chi$ orthogonal to that normal; and a boost tilts the normal and the gradient *together*. Hence Y vanishes in the boosted frame exactly as it does at rest, to all orders in w , and no first-order preferred-frame coupling is generated at all. The clock rate s , being a scalar, is likewise exactly invariant, with none of the second-order time dilation one would naively expect. What remains are two ordinary contributions: the sector's own boosted stress, 6×10^{-16} of the Sun inside Saturn's orbit, and the scalar's leakage into the solar system, governed by the same double-filter screening whose length the static Cassini measurement of γ already bounds. Solving for the screening length each measurement demands gives $\xi > 5.4 \times 10^{-4}$ pc for the preferred-frame bound against $\xi > 8.8 \times 10^{-4}$ pc for γ : the static test is the more demanding, so the preferred-frame bound is satisfied automatically wherever it is, and the programme's existing floor of 0.03 pc clears the binding one by a factor of 34. The contrast with the action that died is sharp and diagnostic: there the clock was coupled to matter directly. No post-Newtonian parameter is extracted from a full metric expansion and none is claimed.

1 Why a preferred frame is usually fatal

Relativistic completions of MOND that introduce a clock field τ , whose level surfaces define a universal simultaneity, acquire a preferred frame. Post-Newtonian theory measures such a frame through α_1 and α_2 , which multiply terms in the metric proportional to the system's velocity w relative to that frame. For the solar system $w \simeq 370 \text{ km s}^{-1}$, so $\beta = w/c = 1.23 \times 10^{-3}$, and the bound $|\alpha_1| \lesssim 10^{-4}$ excludes an order-unity coefficient in front of the tilt by roughly a factor of twelve. The test is not weak.

It has teeth in this programme specifically. An earlier single-metric action [2, 1] carried a cusciton clock and was killed on exactly this test, producing a potential 3×10^{-2} of the Newtonian one at Saturn. The question addressed here is whether that verdict transfers to the sector of [4, 5, 6], in which the same kind of clock makes a dark component cold by driving it onto a critical surface.

2 The action, and the one feature that matters

In signature $(-+++)$ with $c = 1$,

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (R - 2\Lambda) + P(X, \tau) - V(\tau) + sW(Y, \tau) + \gamma X \square \chi \right] + S_r + S_b, \quad (1)$$

with $s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau}$, $n_\mu = -\partial_\mu \tau / s$, $Q = n^\mu \partial_\mu \chi$, $Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \chi \partial_\nu \chi$ and $X = Q^2 - Y$. The feature that decides this paper is that S_r and S_b are *minimally coupled*: matter sees the metric and nothing else. The action that died carried, in addition, an explicit mixing term between clock and scalar which the matter sourced.

3 The sector is blind to the boost

Work in the rest frame of a source moving at velocity w through the clock's frame. There

$$\partial_\mu \tau = s_0 \gamma (1, w, 0, 0), \quad n_\mu = -\partial_\mu \tau / s, \quad (2)$$

so the normal is tilted by w : this is the preferred-frame coupling. Now observe that the background scalar depends only on the clock, $\chi = \bar{\chi}(\tau)$, hence

$$\partial_\mu \bar{\chi} = q \partial_\mu \tau \parallel n_\mu, \quad q \equiv d\bar{\chi}/d\tau. \quad (3)$$

The MOND invariant Y is, by construction, the norm of the part of $\partial\chi$ *orthogonal* to n . A boost tilts n_μ and $\partial_\mu \bar{\chi}$ together, because both are proportional to $\partial_\mu \tau$. Therefore

$$\boxed{Y_{\text{boosted}} = 0 \text{ exactly, to all orders in } w.} \quad (4)$$

This is verified symbolically without expanding in w . The tilt generates no background MOND gradient, so there is no first-order preferred-frame coupling to screen in the first place.

A companion statement is stronger than one expects. Since $s = \sqrt{-\nabla\tau \cdot \nabla\tau}$ is a scalar,

$$s = \sqrt{s_0^2 \gamma^2 (1 - w^2)} = s_0 \quad \text{exactly,} \quad (5)$$

with no dependence on w at all: the second-order time dilation of 1.5×10^{-6} one might expect simply does not occur. Together with (4), the entire background clock sector is blind to the motion, and there is nothing at background level for the preferred frame to act upon.

4 What remains, and which bound binds

Two ordinary contributions survive.

The sector's own boosted stress. The retained sector holds $5.9 \times 10^{-16} M_\odot$ inside Saturn's orbit (and 1.9×10^{-14} inside Neptune's), using the standard local dark matter density scaled by the retained fraction 0.135 that galaxies require [3]. Boosting that stress contributes at the same level: eleven orders below the bound.

The scalar's leakage into the solar system. This is what matter actually feels once the MOND coupling is present, and in this programme it is governed by the double-filter screening [1], with transmission

$$T(x) = 1 - e^{-x} \left(1 + x + \frac{x^2}{2} \right) = \frac{x^3}{6} + O(x^4), \quad x = r/\xi. \quad (6)$$

Because T falls as x^3 , the bound on ξ is a steep function of the tolerance. Solving $T(r/\xi) =$

Table 1: Transmission into Saturn's orbit, $r = 4.62 \times 10^{-5}$ pc.

ξ [pc]	$x = r/\xi$	$T(x)$
5×10^{-4}	9.25×10^{-2}	1.23×10^{-4}
1×10^{-3}	4.63×10^{-2}	1.59×10^{-5}
3×10^{-3}	1.54×10^{-2}	6.04×10^{-7}
1×10^{-2}	4.63×10^{-3}	1.64×10^{-8}
3×10^{-2}	1.54×10^{-3}	6.10×10^{-10}

10^{-4} gives $\xi > 5.4 \times 10^{-4}$ pc for the preferred-frame bound; solving $T(r/\xi) = 2.3 \times 10^{-5}$ gives $\xi > 8.8 \times 10^{-4}$ pc for the static bound on γ . *The static test is the more demanding of the two.* The preferred-frame bound is therefore satisfied automatically wherever γ is, and the programme's existing screening floor of 0.03 pc, set by Cassini, exceeds the binding requirement by a factor of 34 and delivers $T = 6.1 \times 10^{-10}$.

5 Why this action survives where the other died

The contrast is diagnostic rather than incidental. In the action that failed, the clock was coupled to matter directly, so the tilt entered the matter sector at first order and was multiplied by an amplification which that action's own bound forced above 18. In (1) the clock is sourced by the metric and by the scalar, never by the density; the amplification appearing in its linearised clock constraint is $2qd/W = \mu/q$, a ratio of background quantities bounded at order unity by the same margin condition that keeps the kinetic term healthy, spanning 0.05 to 1.8 over the admissible range. A theory can carry a preferred frame and still be invisible to preferred-frame tests, provided the frame reaches matter only through the metric and the sector's own invariant is built orthogonally to it.

6 What is not claimed

No post-Newtonian parameter is extracted from a full metric expansion. What is established is (4) and the exact invariance of s , both symbolically and without expansion in w , together with the two magnitudes above by scaling. The leakage estimate assumes the preferred-frame amplitude is controlled by the same double-filter transmission as the static one; that is the same scalar reaching the same place, but it is an assumption rather than a derivation, and the full expansion should be

carried out. The screening floor of 0.03 pc is this programme’s own. The local dark matter density is the standard 0.4 GeV cm^{-3} scaled by the retained fraction.

7 Reproducibility

`fable_independent_2026/L203_preferred_frame_gate.py` and `L204_boosted_metric_expansion.py` with their committed outputs (commits `b7b8a0ca7`, `c03d2982a`); the matter-budget and static post-Newtonian gates in `L202_budget_and_ppn_gates.py`; the sector’s construction in L192–L201.

References

- [1] The research repository, <https://github.com/carlzimmerman/zimmerman-formula> (2026).
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